

# NexusFinance — OTP

## Costing Analysis

Twilio SMS + Telegram Bot Comparison

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| Document Number | NF-DOC-COST-002         |
| Date            | August 2026             |
| Prepared By     | NDX Digital Support     |
| Classification  | Internal & Confidential |

OTP • Twilio • SMS • Telegram • Costing

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# 1. Executive Summary

This document analyzes the cost of OTP (One-Time Password) delivery for NexusFinance user verification. Two channels are evaluated: Twilio SMS and Telegram Bot API.

## Key Finding

Twilio SMS costs \$0.0105 per OTP (Cambodia). Telegram Bot API is completely FREE. A hybrid strategy — Telegram for linked users, SMS for unlinked — can reduce OTP costs by 60-90%.

# 2. OTP Delivery Channels

| Channel      | Cost Per OTP  | Delivery Speed | Reliability | Coverage            |
|--------------|---------------|----------------|-------------|---------------------|
| Twilio SMS   | \$0.0105      | 1-5 seconds    | 99.5%       | All mobile users    |
| Telegram Bot | \$0.00 (FREE) | Instant        | 99.9%       | Telegram users only |
| Brevo Email  | \$0.00 (FREE) | 5-30 seconds   | 98%         | Email users         |

Note: Telegram Bot API has no per-message cost. The bot is free to use. SMS is the only paid channel.

# 3. Twilio SMS Pricing Breakdown

## 3.1 Per-Message Cost Components

| Component                | Cost      | Notes                                          |
|--------------------------|-----------|------------------------------------------------|
| Base SMS rate (Cambodia) | \$0.0075  | Outbound SMS to Cambodia (+855)                |
| Carrier surcharge        | \$0.0030  | Pass-through fee from local carriers           |
| Long code number rental  | \$1.15/mo | Dedicated phone number for OTP                 |
| Failed message fee       | \$0.0010  | Only charged on delivery failure               |
| Inbound SMS              | \$0.0075  | If using two-way SMS (not recommended for OTP) |

## 3.2 Effective Cost Calculation

For a standard OTP message (160 characters, single segment):

| Line Item              | Cost                        |
|------------------------|-----------------------------|
| Base SMS rate          | \$0.0075                    |
| Carrier surcharge      | \$0.0030                    |
| Number rental (spread) | \$0.0012 (at 1,000 msgs/mo) |
| TOTAL per OTP          | \$0.0117                    |

### Cambodia-Specific Note

Twilio's Cambodia SMS rate is \$0.0075/message — one of the lowest in Southeast Asia. For comparison: Thailand is \$0.025, Vietnam is \$0.015, Philippines is \$0.012. Cambodia benefits from competitive carrier agreements.

## 3.3 Twilio Volume Discounts

| Monthly Volume | Discount           | Effective Rate/OTP |
|----------------|--------------------|--------------------|
| 0 - 100K       | 0%                 | \$0.0105           |
| 100K - 500K    | Up to 15%          | \$0.0089           |
| 500K - 1M      | Up to 25%          | \$0.0079           |
| 1M+            | Custom negotiation | Contact Twilio     |

## 4. Telegram Bot (FREE)

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The Telegram Bot API is completely free for sending messages. There are no per-message fees, no monthly subscriptions, and no volume limits.

### 4.1 Telegram Bot Advantages

- Zero cost per OTP — no per-message charges
- Instant delivery — typically under 1 second
- Rich formatting — Markdown support for OTP messages
- No phone number required — works via Telegram chat ID
- No carrier surcharges or hidden fees
- 99.9% uptime — hosted on Telegram's infrastructure

### 4.2 Telegram Bot Limitations

- Only works for users who have Telegram installed
- Requires user to link their Telegram account first
- Not suitable for users who don't use Telegram
- Bot must be configured and maintained

#### **NexusFinance Integration**

NexusFinance already has a Telegram bot (@nexusfinancefintech\_bot) with phone number sharing and account linking. Users who link their Telegram can receive OTPs via the bot at zero cost.

## 5. Volume Scenarios & Projections

Monthly OTP volume projections for NexusFinance:

### 5.1 Volume Assumptions

| Scenario    | New Users/Mo | Logins/Mo | Password Resets/Mo | Total OTPs/Mo |
|-------------|--------------|-----------|--------------------|---------------|
| Early Stage | 20           | 50        | 5                  | 75            |
| Growth      | 50           | 200       | 15                 | 265           |
| Mature      | 100          | 500       | 30                 | 630           |
| Scale       | 300          | 1,500     | 80                 | 1,880         |
| Enterprise  | 1,000        | 5,000     | 200                | 6,200         |

### 5.2 SMS-Only Cost (100% SMS)

| Volume     | SMS Cost | Number Rental | Total/Mo | Total/Year |
|------------|----------|---------------|----------|------------|
| 75 OTPs    | \$0.79   | \$1.15        | \$1.94   | \$23.28    |
| 265 OTPs   | \$2.78   | \$1.15        | \$3.93   | \$47.16    |
| 630 OTPs   | \$6.62   | \$1.15        | \$7.77   | \$93.24    |
| 1,880 OTPs | \$19.74  | \$1.15        | \$20.89  | \$250.68   |

### 5.3 Telegram-Only Cost (0% SMS)

| Volume     | Telegram Cost | Total/Mo | Total/Year |
|------------|---------------|----------|------------|
| 75 OTPs    | \$0.00        | \$0.00   | \$0.00     |
| 265 OTPs   | \$0.00        | \$0.00   | \$0.00     |
| 630 OTPs   | \$0.00        | \$0.00   | \$0.00     |
| 1,880 OTPs | \$0.00        | \$0.00   | \$0.00     |

## 6. Hybrid Strategy

The recommended approach is a hybrid strategy: Telegram for linked users, SMS for unlinked users. This maximizes cost savings while maintaining coverage.

### 6.1 Hybrid Cost Model

| Scenario     | Telegram % | SMS % | Volume | Cost/Mo | Savings vs SMS-Only |
|--------------|------------|-------|--------|---------|---------------------|
| 10% Telegram | 10%        | 90%   | 630    | \$7.14  | 8%                  |
| 30% Telegram | 30%        | 70%   | 630    | \$5.85  | 25%                 |
| 50% Telegram | 50%        | 50%   | 630    | \$4.56  | 41%                 |
| 70% Telegram | 70%        | 30%   | 630    | \$3.27  | 58%                 |
| 90% Telegram | 90%        | 10%   | 630    | \$1.98  | 75%                 |

#### Realistic Adoption

In Cambodia, Telegram penetration is ~40% among smartphone users. With the NexusFinance bot's phone sharing feature, we can expect 30-50% of users to link their Telegram accounts within the first 3 months.

### 6.2 Recommended Hybrid Mix

| Metric                           | Value        |
|----------------------------------|--------------|
| Target Telegram adoption         | 40% of users |
| SMS fallback                     | 60% of users |
| Cost per 1,000 OTPs (hybrid)     | \$6.30       |
| Cost per 1,000 OTPs (SMS-only)   | \$10.50      |
| Annual savings at 1,000 OTPs/mo  | \$50.40      |
| Annual savings at 10,000 OTPs/mo | \$504.00     |

## 7. Annual Cost Comparison

Year 1 cost comparison across all strategies:

| Strategy                  | 75<br>OTPs/mo | 265<br>OTPs/mo | 630<br>OTPs/mo | 1,880<br>OTPs/mo | 6,200<br>OTPs/mo |
|---------------------------|---------------|----------------|----------------|------------------|------------------|
| SMS-Only                  | \$23.28       | \$47.16        | \$93.24        | \$250.68         | \$795.00         |
| 40% Telegram / 60%<br>SMS | \$14.58       | \$30.30        | \$59.22        | \$159.24         | \$504.00         |
| Telegram-Only             | \$0.00        | \$0.00         | \$0.00         | \$0.00           | \$0.00           |

### 7.1 Cost Per OTP Summary

| Strategy       | Cost/OTP | Annual Cost (1K OTPs/mo) |
|----------------|----------|--------------------------|
| SMS-Only       | \$0.0105 | \$126.00                 |
| Hybrid (40/60) | \$0.0063 | \$75.60                  |
| Telegram-Only  | \$0.00   | \$0.00                   |

## 8. Recommendation

| Criterion          | Recommendation        | Rationale                                   |
|--------------------|-----------------------|---------------------------------------------|
| Cost Optimization  | Hybrid (40% Telegram) | 60% savings vs SMS-only, realistic adoption |
| Coverage           | Twilio SMS            | All users covered, reliable delivery        |
| User Experience    | Telegram Bot          | Instant delivery, rich formatting           |
| Long-term Strategy | Push Telegram linking | Zero cost, better UX                        |

### Final Recommendation

Implement a hybrid OTP strategy: Telegram Bot for linked users (FREE), Twilio SMS for unlinked users (\$0.0105/OTP). With 40% Telegram adoption, the effective cost drops to \$0.0063/OTP — saving 40% vs SMS-only. Promote Telegram linking through the bot's /link command to increase adoption.